

1 Absolute values and discrete valuations

1.1 Introduction

At its core, number theory starts with the ring $\mathbb{Z}$. By the fundamental theorem of arithmetic, every element of $\mathbb{Z}$ can be written uniquely as a product of primes (up to multiplication by a unit ± 1), so it is natural to focus on the prime elements of $\mathbb{Z}$. If p is a prime, the ideal $(p) := p\mathbb{Z}$ is a maximal ideal ($\mathbb{Z}$ has Krull dimension one), and the residue field $\mathbb{Z}/p\mathbb{Z}$ is the finite field $\mathbb{F}_p$ with p elements. The fraction field of $\mathbb{Z}$ is the field $\mathbb{Q}$ of rational numbers. The field $\mathbb{Q}$ and the finite fields $\mathbb{F}_p$ together make up the prime fields: every field k contains exactly one of them, according to its characteristic: k has characteristic zero if and only if it contains $\mathbb{Q}$, and k has characteristic p if and only if k contains $\mathbb{F}_p$.

One can also consider finite extensions of $\mathbb{Q}$, such as the field $\mathbb{Q}(i) := \mathbb{Q}[x]/(x^2+1)$. These are called *number fields*, and each can be constructed as the quotient of the polynomial ring $\mathbb{Q}[x]$ by one of its maximal ideals; the ring $\mathbb{Q}[x]$ is a principal ideal domain and its maximal ideals can all be written as (f) for some monic irreducible $f \in \mathbb{Z}[x]$.

Number fields are one of two types of *global fields* the others are *global function fields*. Let $\mathbb{F}_q$ denote the field with q elements, where q is any prime power. The polynomial ring $\mathbb{F}_q[t]$ has much in common with the integer ring $\mathbb{Z}$. Like $\mathbb{Z}$, it is a principal ideal domain of dimension one, and the residue fields $\mathbb{F}_q[t]/(f)$ one obtains by taking the quotient by a maximal ideal (f) , where $f \in \mathbb{F}_q[t]$ is any irreducible polynomial, are finite fields $\mathbb{F}_{q^d}$, where d is the degree of f . In contrast to the situation with $\mathbb{Z}$, the residue fields of $\mathbb{F}_q[t]$ all have the same characteristic as its fraction field $\mathbb{F}_q(t)$, which plays a role analogous to $\mathbb{Q}$. Global function fields are finite extensions of $\mathbb{F}_q(t)$.

Associated to each global field k is an infinite collection of *local fields* corresponding to the completions of k with respect to its absolute values; when $k = \mathbb{Q}$, these completions are the field of real numbers $\mathbb{R}$ and the p -adic fields $\mathbb{Q}_p$ (as you will prove on Problem Set 1).

The ring $\mathbb{Z}$ is a principal ideal domain (PID), as is $\mathbb{F}_q[t]$, and in such fields every nonzero prime ideal is maximal and thus has an associated *residue field*. For both $\mathbb{Z}$ and $\mathbb{F}_q[t]$ these residue fields are finite, but the characteristics of the residue fields of $\mathbb{Z}$ are all different (and distinct from the characteristic of its fraction field), while those of $\mathbb{F}_q[t]$ are all the same.

Remark 1.1. All rings have a multiplicative identity that is preserved by ring homomorphisms. Except where noted otherwise, the rings we shall consider are all commutative.

1.2 Absolute values

A reference for much of this material in this section is [3, Chapter 1].

Definition 1.2. An *absolute value* on a field k is a map $|\cdot|: k \rightarrow \mathbb{R}_{\geq 0}$ such that for all $x, y \in k$ the following hold:

1. $|x| = 0$ if and only if $x = 0$;
2. $|xy| = |x||y|$;
3. $|x + y| \leq |x| + |y|$.

If the stronger condition

4. $|x + y| \leq \max(|x|, |y|)$

also holds, then the absolute value is *nonarchimedean*; otherwise it is *archimedean*.